INCH-POUND

MIL-P-44311B
3 September 1993
SUPERSEDING
MIL-P-44311A
30 March 1990

MILITARY SPECIFICATION

# PORK SAUSAGE LINKS IN BRINE, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers pork sausage links in brine, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8905

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SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |

MILITARY

| | | |
|---|---|---|
| MIL-L-1497 | - | Labeling of Metal Cans for Subsistence Items |
| MIL-L-35078 | - | Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For |
| MIL-C-44340 | - | Can, Tray Pack |

STANDARDS

MILITARY

| | | |
|---|---|---|
| MIL-STD-105 | - | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | - | Marking for Shipment and Storage |
| MIL-STD-900 | - | Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)



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United States Standards for Condition of Food Containers

(Copies are available from the Chairperson, Condition of Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2506, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARIMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Applications for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-Sensitive Tape

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

(Application for copies should be addressed to the AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)



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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Pork. The pork shall be such that the sausage links will conform to the requirements specified in 3.6.

3.2.2 Water. Water used for formulation and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.3.1 Salt tablets. Salt tablets shall be noniodized, white, refined sodium chloride and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.4 Dextrose. Dextrose shall comply with the Food Chemicals Codex.

3.2.5 Sugar, white, granulated. Sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.6 Pepper, white, ground. Ground white pepper shall be derived from the dried mature berries of Piper nigrum L. from which the outer covering or the outer and inner covering have been removed. The ground pepper shall have a characteristic, penetrating odor, a hot biting, pungent flavor, and a light color. The ground white pepper shall contain not less than 1.0 mL of volatile oil per 100 grams of ground white pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.7 Sage, ground. Ground sage shall be derived from the dried leaves of Salvia officinalis L. The ground sage shall be green to grayish-green in color and shall possess a strong, fragrant, aromatic odor, and a slightly



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bitter taste. The ground sage shall contain not less than 1.5 mL of volatile oil per 100 grams of ground sage and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 20 sieve.

3.2.8 Pepper, black, ground. Ground black pepper shall have been derived and ground from deep brown to black, deep-set, wrinkled immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting, pungent flavor, and a light gray to speckled black-gray color. The ground black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.9 Ginger, ground. Ground ginger shall be derived from the root or rhizome of Zingiber officinale Roscoe and shall possess an aromatic odor and flavor with a tan to pale brown color. The ground ginger shall contain not less than 1.5 mL of volatile oil per 100 grams of ground ginger and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 35 sieve.

3.2.10 Sausage casing. Sausage casing shall be made of reconstituted collagen (edible) designed for processing (see 6.4) and shall be fabricated to maintain integrity during processing.

3.2.11 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Pork sausage link preparation. The pork sausage links shall be formulated and prepared as follows:

| Ingredients | Percent by weight |
|---|---|
| Pork, ground 1/ | 93.43 |
| Water or ice | 2.80 |
| Salt 2/ | 1.64 |
| Dextrose | 0.82 |
| Sugar, white, granulated | 0.82 |
| Pepper, white, ground | 0.23 |
| Sage, ground | 0.18 |
| Pepper, black, ground | 0.06 |
| Ginger, ground | 0.02 |

1/ The pork shall be chopped or ground to a moderately or slightly coarse texture.



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2/ The total amount of salt in the formula may be adjusted, as necessary, to produce a product that complies with the finished product salt requirements (see 3.6).

a. The sausage mixture shall be stuffed into collagen casings of a size to ensure compliance with the finished product requirements (see 3.6).

Skinless sausage links of a size to ensure compliance with the finished product requirements (see 3.6).

b. Prior to stuffing into collagen casings or mechanically forming, the sausage mixture shall be held for not more than 36 hours in the temperature range of 28°F to 40°F.

c. If not immediately cooked after mechanically forming or stuffing into casings, the sausage links shall be packed into USDA approved moisture and oxygen impermeable material. The packaged, raw sausage links shall be maintained at an internal temperature of 28°F to 40°F for not more than four days prior to cooking. If destined for freezing, the packaged, raw sausage links shall be frozen to 0°F or below within 72 hours after cooking and held at an internal temperature of 0°F or below for not more than 15 days prior to cooking.

d. The sausage links shall be oven-cooked until no pink or red color remains in the links. In addition, the cooking procedure shall be sufficient to produce a light brown to dark brown surface color.

e. If not filled in the tray pack cans and sealed (see 3.4) immediately after cooking, the cooked sausage links shall be packaged into USDA approved moisture and oxygen impermeable material. The packaged, cooked sausage links shall be maintained at an internal temperature of 28°F to 40°F for not more than 15 days prior to tray pack can filling. If destined for freezing, the packaged, cooked sausage links shall be frozen to 0°F or below within 72 hours after cooking and held at an internal temperature of 0°F or below for not more than 60 days prior to tray pack can filling and sealing.

3.3.2 Brine preparation. Brine shall be prepared in a manner to ensure compliance with finished product salt requirements. (At the time of filling, salt tablets may be substituted for granular salt in brine preparation.) Brine shall be made fresh daily, and brine remaining at the end of a production day shall be discarded.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with product to conform to the finished product requirements and to the following requirements:

a. Ninety uniformly sized sausage links shall be packed in the tray pack can in two layers. The bottom and top layer shall consist of three rows packed across the width of the tray pack can. Extra links shall be filled into the spaces remaining at the ends of the tray pack can.



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b. The product temperature at the time of filling shall be 28°F to 40°F unless filling of the tray pack cans occurs immediately after cooking.

c. Each tray pack can shall be filled with the product and brine to a total can content weight of 104 ounces. Brine may be filled hot or cold.

d. Immediately after filling, each can shall be sealed in accordance with the can manufacturer's guidelines/requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable (see 4.5.5), and under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6s).

e. Each filled and sealed tray pack can shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value ($F_o$) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, musty, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall have a total can content weight of less than 102 ounces.

e. The average total can content weight shall be not less than 104 ounces.

f. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

g. Drained weight of the 90 sausage links in any individual tray pack can shall be not less than 54.0 ounces.

h. No individual can shall contain less than 90 pork sausage links.

i. Sausage links shall be of uniform diameter and length.

j. Sausage links shall fit into the tray pack can without abnormal distortion or breaking.



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k. There shall be not more than three broken sausage links in a tray pack.

l. Sausage links shall not be soft, mushy, or abnormally distended.

m. For sausage links in collagen casing, the casings shall maintain their integrity (shall not show evidence of sloughing).

n. Links shall have a surface color ranging from light brown to dark brown.

o. Sausage links shall not have an uncooked or raw internal color.

p. The product shall show no evidence of excessive heating (materially darkened or scorched).

q. The average fat content of the finished product shall be not greater than 33.0 percent and no individual tray pack shall have a fat content greater than 35.0 percent.

r. The salt content of any individual tray pack shall be not less than 1.5 percent nor greater than 2.5 percent.

s. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The pork component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under the Meat and Poultry Products Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.



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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material examination. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Pork sausage examination. Examination of pork sausage links shall be made to determine compliance with the requirements of 3.3.1. In addition, records of formulation, temperatures and holding times shall be maintained. When sausage links are obtained from a contractor other than the prime contractor, the links shall receive a source inspection to determine compliance with 3.3.1 and shall be so certified by the USDA. Failure to comply with requirements shall be cause for rejection of the involved product.

4.5.1.2 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC International and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.



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4.5.3 Tray pack inspection. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Total can content weight examination. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average total can content weight is less than 104 ounces or if the total can content weight of any individual can is less than 102 ounces, the lot shall be rejected.

4.5.3.2 Double sampling plan for product examination. The finished product shall be examined for the defects listed in table I utilizing the double sampling plans indicated in MIL-STD-105. The lot size shall be expressed in tray pack cans. The sample unit shall be one filled and sealed tray pack can. The inspection level shall be S-3 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for minor defects. The sample cans shall be heated in accordance with the heating instructions on the can label.

TABLE I. Product defects 1/ 2/

| Category | | Defect |
|---|---|---|
| Major | Minor | |
| 101 | | Presence of a bone piece measuring 0.3 inch or more in any dimension |
| 102 | | Drained weight of 90 sausage links in any individual tray pack can is less than 54.0 ounces 3/ |
| 103 | | Less than 90 pork sausage links in any individual can |
| 104 | | Sausage links not uniform in diameter and in length |
| 105 | | Fit of links in tray pack can causing abnormal distortion or breaking |
| 106 | | More than three broken sausage links per tray pack |
| 107 | | Presence of soft, mushy, or abnormally distended links |



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TABLE I. Product defects 1/ 2/

| Category | | Defect |
|---|---|---|
| Major | Minor | |
| 108 | | Pork sausage links do not have a surface color ranging from light brown to dark brown |
| 109 | | Sausage links possess an uncooked or raw internal color |
| 110 | | Product shows evidence of excessive heating (materially darkened or scorched) |
| | 201 | Sausage links not arranged in a layered manner, with the bottom and top layers consisting of three rows of links packed across the width of the tray pack can |
| | 202 | For sausage links in collagen casing, the casing material shows evidence of sloughing |

1/ The presence of any foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, stale, sour, rancid, musty, moldy), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot.

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a flat-bottom container. A minimum of three times the tray pack can's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated such as to liquefy rendered fat without breaking the links. The contents shall then be poured into the U.S. Standard 1/2 inch sieve in a manner that will distribute the product over the sieve without breaking the links. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at approximately a 45° angle and allowed to drain for two minutes before determining the drained weight. Ninety pork sausage links shall be weighed to determine the drained weight. The drained weight shall be reported to the nearest 0.1 ounce.



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n is required to depth constant at 10 inches. It a stronger cross sectio

4.5.3.3 Fat and salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of the AOAC, methods 976.21, 985.15, or 960.39, and for salt content in accordance with the Official Methods of Analysis of AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened tray pack cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that the inspection for labeling shall be in accordance with 4.5.4.1. In addition, scratches, scuffs, or abrasions that occur on the outside coating as a result of the filling, sealing, and thermoprocessing of the tray cans shall not be scored as a defect.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4) except, for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, for example, label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's guidelines/ requirements and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to $75^{\circ}F \pm 5^{\circ}F$, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made



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when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect, or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and four ounces of product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class



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of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of class domestic, grade 175 fiberboard.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern. Boxes may be stacked by interlocking and reversing each tier, or by columnar stacking with paperboard or fiberboard sheets placed between each tier. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three digit letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (for example, March 19, 1992 would be coded as 2078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used)



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shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be made, stamping is permitted, on one 1001 by 200 end of the can. The labeling shall be legible when examined as specified in 4.5.4.1 after preparation of the product in accordance with heating instructions. Paper labels are not permitted. In addition, cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently 25 to 30 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open in usual manner leaving the loose lid in place. Place in a 350°F oven 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 5 sausages each.

As an alternate labeling method, a preprinted, self-adhering, 0.002-inch thick, clear, polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Levels of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Ingredient information.

6.4.1 Sausage casing. It has been found that casings #210-613 manufactured by Devro, Inc., meets the requirements of 3.2.10 and performs satisfactorily in this product.

6.5 Subject term (key word) listing.

Canned food
Combat field feeding
Food processing
Operational ration
Shelf stable

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DLA - SS

Preparing activity:

Army - GL

(Project 8905-1115)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-P-44311B | 1993 September 3 |

**3. DOCUMENT TITLE**
PORK SAUSAGE LINKS IN BRINE, THERMOSTABILIZED, TRAY PACK

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| U.S. Army Natick RD&E Center | (1) Commercial<br>508-651-4501 | (2) AUTOVON/DSN<br>256-4501 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: SATNC-WRE<br>Natick, MA 01760-5018 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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